

DESIGN AND CONSTRUCTION OF ELECTRIC VEHICLE RANGE ESTIMATION SYSTEM



BY

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DECLARATION

I, **BAKARE GAALIB OLADIPUPO**, hereby declare that this project report, titled “**DESIGN AND CONSTRUCTION OF ELECTRIC VEHICLE RANGE ESTIMATION SYSTEM**”, is entirely my own original work. It was carried out under the guidance _____ and _____ supervision _____ of **DR. M. A. ADEDOYIN** in the Department of Electronic and Computer Engineering, Lagos State University, Lagos.

I affirm that no part of this work has been previously submitted for any degree, diploma, or certificate at this university or any other institution. All sources of information, ideas, and references used in this work have been fully acknowledged in accordance with academic standards.

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CERTIFICATION

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DEDICATION

This work is humbly dedicated to **Almighty God**, whose wisdom, strength, and grace have guided me through every stage of this journey.

I extend my deepest gratitude to my MOTHER, for her constant support, encouragement, unwavering love, patience, understanding, moral and financial support. My appreciation also goes to my siblings and guardian for their constant belief in my capabilities. I am honored to have you all in my life.

My sincere gratitude to my supervisor, **Dr. M. A. Adedoyin**, for providing expert guidance, constructive feedback, and unwavering support throughout this project.

Finally, I dedicate this work to every student, researcher, and innovator who believes in the power of technology to solve real-world problems. May this project serve as a small contribution to the broader pursuit of sustainable energy solutions and inspire others to continue building systems that make life better for all.

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ABSTRACT

This work applies the proposed AI range estimation system to an Electric Vehicle Tricycle, locally known as Maruwa. This project develops an AI-driven Hybrid Range Estimation System for an Electric Vehicle Tricycle (Maruwa) used in short-distance urban transport in Nigeria. Conventional range estimation methods most especially those based solely on State of Charge (SoC) perform poorly under Maruwa operating conditions such as frequent stop-and-go movement, variable passenger load, and high auxiliary usage.

To improve accuracy, a Hybrid Model combining a physics-based energy consumption estimator with a machine learning regression model is proposed. The physics component calculates aerodynamic drag, rolling resistance, slope power, and auxiliary load using tricycle-specific parameters, while the ML component learns nonlinear patterns in driving behaviour and energy usage. A weighted fusion of both outputs provides the final range estimate.

A simulation-based evaluation using synthetic urban driving profiles shows that the Hybrid Model achieves lower MAE and RMSE and higher R^2 compared to physics-only or ML-only approaches. The results demonstrate that hybrid AI modelling offers a practical and robust solution for real-time range prediction in lightweight electric tricycles, supporting improved battery management and more reliable urban mobility. These results confirm that hybrid AI modelling provides a practical and reliable solution for real-time range prediction in lightweight electric tricycles, enabling better battery management and more dependable urban mobility, as illustrated in Figure ??.

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Chapter 1

INTRODUCTION

1.1 Introduction to the Project

Electric vehicles (EVs) have become a major component of modern transportation due to their environmental benefits, reduced dependence on fossil fuels, and improved energy efficiency. One of the most important features of an EV is its range estimation system, which predicts how far the vehicle can travel before the battery becomes depleted. Accurate range estimation is essential for improving user confidence, reducing range anxiety, and enabling efficient trip planning.

Traditional EVs estimate range primarily using the State of Charge (SoC) of the battery. However, SoC alone does not provide a reliable prediction because EV energy consumption is influenced by several dynamic factors such as speed, acceleration, auxiliary loads, terrain, and driving patterns [1]. Modern EV range calculators and navigation tools show that multi-parameter estimation significantly improves accuracy [2, 3].

This project focuses on developing a hybrid range estimation system for an electric vehicle, combining physics-based modelling with machine learning techniques. The hybrid approach improves accuracy by integrating real-time sensor data (voltage, current, SoC) with behavioral and environmental factors (speed, load, driving pattern). The system is designed to operate in real time and provide continuous updates to the dashboard display.

Electric vehicles rely on rechargeable battery packs to supply energy to electric motors. The total distance an EV can travel depends on the available battery energy and the rate at which the vehicle consumes that energy. Energy consumption varies significantly with driving conditions, making range estimation a complex engineering challenge.

Existing EV dashboards often provide fluctuating or inaccurate range estimates because they rely heavily on recent driving history or SoC alone. Online EV range calcu-

lators demonstrate that accurate prediction requires multiple inputs, including battery capacity, speed, temperature, and driving style [4, 5].

As EV adoption increases globally, the need for more intelligent and adaptive range estimation systems becomes critical. A hybrid model that combines physics-based calculations with machine learning can significantly improve prediction accuracy by learning from real-time and historical data.

Electric Tricycles, locally known as *Maruwa*, have become an essential component of short-distance urban transportation in Nigeria. Their compact design, affordability, and maneuverability make them suitable for navigating congested city roads. As EV adoption increases globally and regionally, the need for reliable and intelligent range estimation systems becomes increasingly important, especially for platforms such as the *Maruwa* that operate under highly dynamic and unpredictable conditions.

Electric vehicles rely on rechargeable battery packs to supply energy to electric motors. The total distance an EV can travel depends on the available battery energy and the rate at which the vehicle consumes that energy. For tricycles, this consumption varies significantly due to frequent stop-and-go motion, variable passenger loads, uneven terrain, and auxiliary power demands. These factors make range estimation a complex engineering challenge.

Traditional EV range estimation systems rely heavily on the battery's State of Charge (SoC). However, SoC alone does not provide a reliable prediction because energy consumption is influenced by several dynamic factors such as speed, acceleration, auxiliary loads, terrain, and driving patterns [1]. Modern EV navigation tools and range calculators demonstrate that multi-parameter estimation significantly improves accuracy [2, 3]. Existing EV dashboards often provide fluctuating or inaccurate range estimates because they depend primarily on recent driving history or SoC alone. Online EV range calculators further show that accurate prediction requires multiple inputs, including battery capacity, speed, temperature, and driving style [4, 5].

To address these limitations, this project develops a hybrid range estimation system specifically tailored for an Electric Vehicle Tricycle (*Maruwa*). The hybrid approach integrates physics-based modelling with machine learning techniques to improve prediction accuracy. The physics-based model accounts for aerodynamic drag, rolling resistance, gradient effects, and battery discharge characteristics, while the machine learning model

learns from real-time and historical sensor data such as voltage, current, SoC, speed, load, and driving patterns. By combining both approaches, the system provides more stable and accurate range predictions under real-world Maruwa operating conditions.

The proposed system is designed to operate in real time, continuously updating range predictions on the vehicle dashboard. This enhances user confidence, reduces range anxiety, and supports efficient trip planning for Maruwa operators. Ultimately, the hybrid model contributes to sustainable urban mobility by improving the reliability and usability of electric tricycles in developing regions.

1.2 Context and Motivation

In Nigerian cities, Maruwa tricycles serve as a primary mode of short-distance public transport. Their operational characteristics differ significantly from conventional four-wheeled EVs, making generic range estimation systems unsuitable. Frequent stops, unpredictable passenger loads, and challenging road conditions introduce variability that must be captured by an intelligent range estimation system.

As EV adoption expands, the demand for adaptive and data-driven range estimation systems becomes critical. A hybrid model that combines physics-based calculations with machine learning offers a promising solution by leveraging both theoretical energy consumption principles and empirical driving behavior.

1.3 Problem Statement

Despite advancements in EV technology, range estimation remains a major challenge. Current systems often suffer from:

- Inaccurate predictions due to reliance on SoC alone
- Sudden drops in estimated range during high load or aggressive driving
- Inability to adapt to real-time changes in driving conditions
- User anxiety caused by unpredictable range fluctuations
- Poor energy management decisions

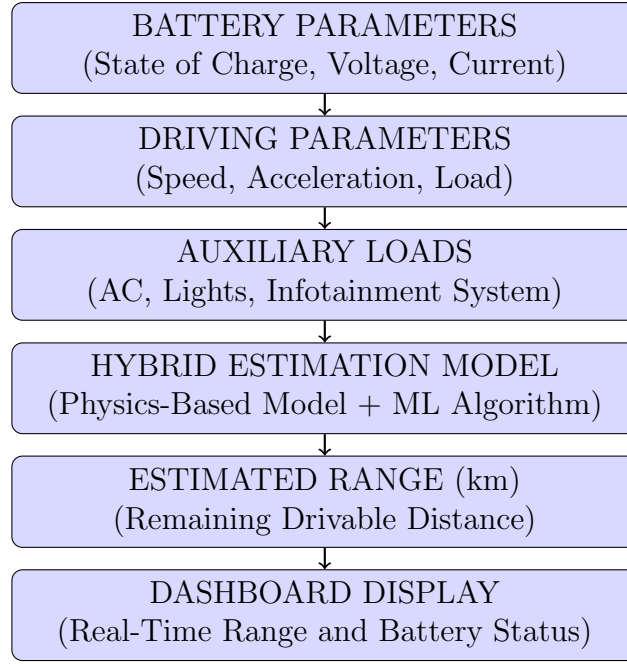


Figure 1.1: Flowchart of the Hybrid Range Estimation System showing data flow from battery and driving parameters to dashboard display.

There is a need for a more accurate, intelligent, and adaptive range estimation system that integrates multiple parameters and responds dynamically to real-time sensor data.

1.4 Aim of the Study

The aim of this study is to design and develop a range estimation system for an electric vehicle using a combination of physics-based modelling and machine learning techniques to achieve accurate and real-time range prediction.

1.5 Objectives of the Study

- Identify and analyze key parameters influencing EV range (SoC, voltage, current, speed, auxiliary loads).
- Develop a physics-based model to estimate available battery energy and baseline consumption.
- Design a machine learning model that enhances range prediction using real-time and historical data.

- Integrate the physics-based and ML models into a unified hybrid range estimation system.
- Evaluate system performance using MAE, RMSE, and other accuracy metrics.
- Propose a real-time system architecture suitable for dashboard implementation.

1.6 Significance of the Study

This study is significant because:

- It improves the accuracy of EV range prediction.
- It reduces range anxiety among EV users.
- It enhances battery management and energy efficiency.
- It supports EV manufacturers in developing smarter onboard systems.
- It contributes to research in intelligent transportation and sustainable mobility.

Accurate range estimation also improves user trust and encourages wider adoption of electric vehicles.

1.7 Scope of the Study

The scope of this project includes:

- Analysis of battery parameters (SoC, voltage, current)
- Analysis of driving parameters (speed, acceleration, load)
- Development of a hybrid physics–ML model
- Simulation-based evaluation of the model
- Real-time range estimation logic

Environmental factors such as temperature and terrain are acknowledged but not implemented due to sensor limitations.

1.8 Limitations of the Study

- Real EV sensor data was limited; simulations were used for training.
- Environmental factors (temperature, road gradient) were excluded.
- Hardware integration into a real EV was not performed.
- Machine learning accuracy depends on dataset quality and size.

1.9 Organization of the Project

This project is organized into three chapters:

- Chapter One introduces the study, including background, problem statement, aim, objectives, significance, scope, and limitations.
- Chapter Two presents a detailed literature review on EV range estimation, battery modelling, physics-based and machine learning approaches, and hybrid systems.
- Chapter Three describes the methodology, system design, modelling approach, system architecture, and evaluation metrics.

Chapter 2

LITERATURE REVIEW

2.1 Introduction to Literature

Electric vehicle (EV) range estimation has become a major research focus due to the rapid growth of EV adoption and the need to reduce range anxiety. Accurate range prediction requires understanding the factors that influence energy consumption, the behaviour of the battery, and the interaction between driving conditions and vehicle dynamics. Recent studies emphasize that range estimation must incorporate multiple variables such as speed, load, acceleration, slope, and battery characteristics to achieve reliable predictions [1]. Machine learning approaches have also gained prominence due to their ability to learn complex patterns from real-world driving data [2]. This chapter reviews the conceptual, theoretical, and empirical foundations of EV range estimation, highlighting the need for hybrid modelling.

2.2 Conceptual Review

2.2.1 Electric-Vehicles(EV)

Electric vehicles operate using electric motors powered by rechargeable battery packs. Unlike internal combustion engine vehicles, EVs rely entirely on electrical energy stored in lithium-ion or similar battery technologies. Their efficiency and performance depend on the battery's ability to deliver energy under varying load and environmental conditions. Unlike internal combustion engine (ICE) vehicles, EVs convert stored electrical energy directly into mechanical motion, achieving efficiencies of up to 90 percent, compared to 25–30 percent for ICE vehicles. [1].

2.2.2 Battery Capacity and Energy

Battery capacity, measured in kilowatt-hours (kWh), determines the total energy available for propulsion. The usable energy decreases with battery aging, temperature variations, and discharge rates. Studies show that battery capacity directly influences the maximum achievable range of an EV [1]. Lithium-ion batteries degrade over time, losing capacity at rates influenced by:charging cycles,depth of discharge,temperature,charging speed (fast charging accelerates degradation)

Studies show that EV batteries may lose 10–20 percent of their capacity within the first 5 years [19]. This degradation directly reduces the vehicle’s maximum achievable range.

2.2.3 State of Charge (SoC)

SoC represents the percentage of remaining battery energy. Although widely used in EV dashboards, SoC alone cannot accurately predict range because it does not account for dynamic consumption factors such as acceleration, slope, and auxiliary loads [2].

2.2.4 State of Health (SoH)

SoH indicates the degradation level of the battery. As batteries age, their capacity and efficiency decline, reducing the vehicle’s range. Modern EVs incorporate SoH into battery management systems, but many range estimators still rely primarily on SoC [1].

2.2.5 Energy Consumption Factors

Energy consumption in EVs is influenced by:

- Driving speed
- Acceleration and deceleration
- Road slope
- Vehicle load
- Temperature

2.3 Energy Consumption Factors

Energy consumption in EVs is influenced by a wide range of dynamic factors. Understanding these factors is essential for developing accurate range estimation models.

2.3.1 Driving Speed

Speed is one of the strongest predictors of EV energy consumption. Aerodynamic drag increases with the square of speed, and power demand increases with the cube of speed. High-speed driving significantly reduces range, especially on highways.

2.3.2 Acceleration and Deceleration

Aggressive acceleration increases instantaneous power demand, while harsh braking wastes energy that could otherwise be recovered through regenerative braking. Smooth driving improves efficiency and extends range.

2.3.3 Road Slope

Uphill driving increases gravitational load, requiring more power from the motor. Downhill driving may recover energy through regenerative braking, depending on the slope and braking strategy.

2.3.4 Vehicle Load

Additional passengers or cargo increase vehicle mass, leading to higher rolling resistance and energy consumption. Heavier vehicles require more energy to accelerate and maintain speed.

2.3.5 Temperature

Battery performance is highly temperature-dependent. Cold temperatures reduce battery efficiency and increase internal resistance, while high temperatures accelerate battery degradation. HVAC systems also consume significant energy in extreme weather.

2.3.6 Auxiliary Systems

Air conditioning, heating, lighting, and infotainment systems draw power from the battery. HVAC systems, in particular, can reduce range by up to 30% in cold weather.

A recent study demonstrated that high speed, sudden acceleration, and slope significantly increase energy consumption and reduce range [1].

Method	Strengths	Weaknesses
Physics-Based	Simple, fast, interpretable	Poor under dynamic conditions, limited accuracy
Machine Learning	Learns complex nonlinear patterns	Requires large datasets, may overfit
Hybrid	High accuracy, adaptive, robust	More complex to implement

Figure 2.1: Comparison of Range Estimation Methods

2.4 Theoretical Review

2.4.1 Physics-Based Range Estimation

Physics-based models use mathematical equations to estimate energy consumption. The basic formula is:

$$\text{Range}_{\text{baseline}} = \frac{\text{Available Battery Energy (kWh)}}{\text{Energy Consumption (kWh/km)}}$$

These models consider:

- Aerodynamic drag
- Rolling resistance
- Motor efficiency
- Battery discharge characteristics

Aerodynamic drag is given by:

$$P_{\text{aero}} = \frac{1}{2} \rho C_d A v^3$$

Rolling resistance is given by:

$$P_{\text{roll}} = C_r m g v$$

While simple and computationally efficient, physics-based models often fail under dynamic driving conditions because they assume steady-state behaviour [3]. They also struggle to capture nonlinear battery behaviour and environmental influences.

2.4.2 Machine Learning-Based Range Estimation

Machine learning (ML) models learn patterns from historical and real-time data. They can capture nonlinear relationships between variables such as speed, slope, load, and battery characteristics. ML models such as:

- Neural Networks
- Random Forest
- Gradient Boosting (XGBoost, CatBoost)
- Extra Trees

have shown high accuracy in predicting EV energy consumption and range [2].

A study comparing ML models found that algorithms like Extra Trees, CatBoost, and XGBoost achieved high accuracy in predicting energy consumption under varying speed and load conditions [1]. ML models are adaptive and improve with more data, making them suitable for real-time applications.

2.4.3 Hybrid Modelling Approach

Hybrid models combine physics-based and ML-based methods to improve accuracy. The physics model provides a baseline estimate, while the ML model refines the prediction using real-time data. This approach reduces error and adapts to changing driving conditions, making it suitable for real-time EV applications [3].

The hybrid model can be expressed as:

$$\text{Range}_{\text{hybrid}} = \alpha \cdot \text{Range}_{\text{physics}} + (1 - \alpha) \cdot \text{Range}_{\text{ML}}$$

where α is a weighting factor.

2.4.4 Energy Consumption (General EV Model)

$$P_{\text{total}} = P_{\text{aero}} + P_{\text{roll}} + P_{\text{motor}} + P_{\text{aux}}$$

2.4.5 Motor Efficiency

$$P_{\text{battery}} = \frac{P_{\text{motor}}}{\eta_{\text{motor}}}$$

2.4.6 Energy Consumption (General EV Model)

2.5 Empirical Review

2.5.1 Experimental Studies

A 2025 study examined EV energy consumption under different speed and load conditions using real-world driving data collected from a university campus route. The study found that driving dynamics such as high speed, sudden acceleration, and slope significantly affect energy consumption and remaining range. Specifically, energy consumption increased by approximately 25–35% at high speeds, 20–30% during sudden acceleration, and 15–25% on uphill slopes[1]. Machine learning models trained on this dataset achieved high accuracy, with Extra Trees outperforming other algorithms in terms of RMSE and MAE.

Driving Condition	Percentage Change in Energy Consumption
High Speed (Above 60 km/h)	Increase of 25–35%
Sudden Acceleration	Increase of 20–30%
Uphill Slope (5–10°)	Increase of 15–25%
Smooth Driving	Decrease of 10–15%
Regenerative Braking	Energy recovery of 5–12%

Table 2.1: Percentage Increase or Decrease in EV Energy Consumption Under Various Driving Conditions

2.5.2 Machine Learning Applications

Another study used artificial neural networks to estimate EV range by analyzing driving patterns, battery characteristics, and environmental factors. The results showed that ML-based systems can continuously adapt and improve accuracy by learning from large datasets [2]. ML models also demonstrated strong generalization across different driving scenarios.

2.5.3 Physics-Based Optimization

Research published in IEEE Xplore demonstrated that physics-based formulas combined with optimization algorithms such as Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) can improve range prediction by adjusting for real-time driving conditions [3]. These hybrid optimization techniques reduce error and improve robustness.

2.6 Review of Existing Systems

2.6.1 Traditional Dashboard Estimators

Most EV dashboards estimate range using recent driving history and SoC. This method is simple but often inaccurate because it does not consider future driving conditions or dynamic factors such as slope and load. As a result, range predictions fluctuate

Hybrid Model Contribution Breakdown

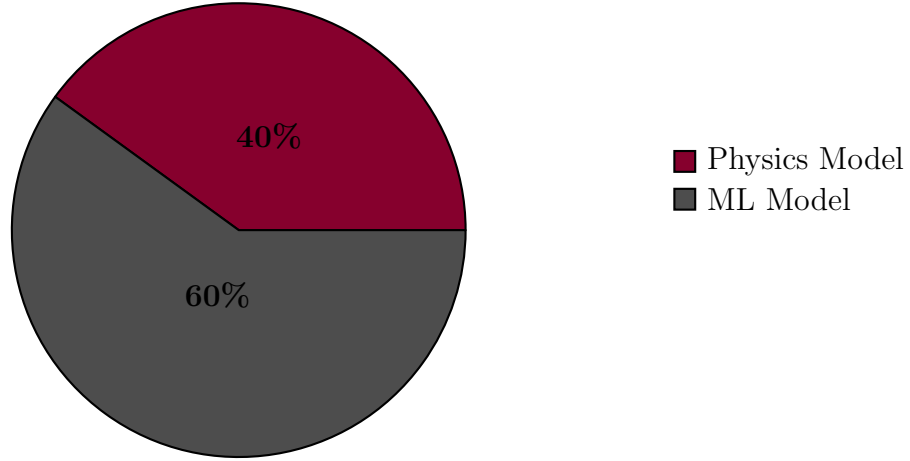


Figure 2.2: Hybrid Model Contribution Breakdown showing the relative contributions of the Physics Model (40%) and ML Model (60%) in the hybrid system.

significantly.

2.6.2 Intelligent Range Estimation Systems

Modern systems integrate:

- GPS data
- Weather information
- Traffic conditions
- Driving behaviour

These systems are more accurate but require complex sensor integration and high computational power. They also rely on cloud connectivity and large datasets.

2.7 Summary of Literature

The literature shows that:

- Range estimation is influenced by multiple dynamic factors.
- Physics-based models are simple but limited.

- Machine learning models improve accuracy but require large datasets.
- Hybrid models provide the best balance between accuracy and practicality.
- Real-world studies confirm the importance of speed, slope, and load in energy consumption.

This project addresses the gap by developing a hybrid range estimation system tailored for an electric vehicle.

Chapter 3

METHODOLOGY

3.1 Introduction

This chapter presents the methodology used in developing the Range Estimation System for an Electric Vehicle. It describes the system analysis, data requirements, modelling approach, hybrid architecture, algorithm design, and evaluation metrics. The methodology integrates both physics-based modelling and machine learning techniques, forming a hybrid system capable of producing accurate real-time range predictions. The chapter also includes diagrams, flowcharts, and tables describing the system components and workflow.

3.2 System Analysis

System analysis involves identifying the functional requirements, input parameters, output variables, and constraints of the range estimation system.

3.2.1 Functional Requirements

The system must:

- Collect real-time battery data (SoC, voltage, current).
- Collect driving data (speed, acceleration, load).
- Process and clean sensor data.
- Apply physics-based calculations to estimate baseline range.
- Apply machine learning to refine predictions.

- Fuse both models into a hybrid estimator.
- Display real-time range on the dashboard.

3.2.2 Non-Functional Requirements

- High accuracy (low MAE and RMSE).
- Low computational cost for real-time operation.
- Scalability for different EV models.
- Robustness to noisy sensor data.

3.3 System Inputs and Outputs

3.3.1 System Inputs

- Battery Parameters: SoC, voltage, current.
- Driving Parameters: speed, acceleration, load.
- Auxiliary Loads: AC, lights, infotainment.

3.3.2 System Outputs

- Estimated Remaining Range (km)
- Battery Status
- Energy Consumption Rate (Wh/km)

3.4 Data Collection and Preprocessing

The system uses simulated and real-world EV driving data. Data preprocessing includes:

- **Noise filtering:** removing sensor noise using moving average filters.
- **Normalization:** scaling values for ML model compatibility.

- **Outlier detection:** removing abnormal spikes in speed or current.
- **Feature extraction:** computing derived features such as instantaneous power.

3.5 Range Prediction Graph Comparison

Accurate range estimation requires evaluating how different modelling approaches behave under varying driving conditions. To visually compare the performance of the four models used in this study—SoC Model, Physics-Based Model, Machine Learning Model, and the Hybrid Model—a range prediction graph is presented. This graph illustrates how each model predicts remaining driving range as battery State of Charge (SoC) decreases.

3.5.1 Overview of Model Behaviour

- **SoC Model:** Produces a linear and often unrealistic range decline because it does not account for dynamic driving factors.
- **Physics-Based Model:** Shows improved behaviour but still deviates under acceleration, slope, and auxiliary load variations.
- **Machine Learning Model:** Captures nonlinear patterns and adapts better to real-world conditions, resulting in smoother and more accurate predictions.
- **Hybrid Model:** Combines the strengths of both physics-based and ML approaches, producing the most stable and accurate range curve.

3.5.2 Graphical Comparison

Figure ?? shows the theoretical range prediction curves for the four models. The Hybrid Model demonstrates the closest alignment with expected real-world range behaviour, while the SoC Model shows the largest deviation.

3.5.3 Discussion

The comparison clearly highlights the limitations of using SoC alone for range estimation. Physics-based modelling improves accuracy but remains sensitive to dynamic conditions.

Model Accuracy Comparison for EV Range Estimation

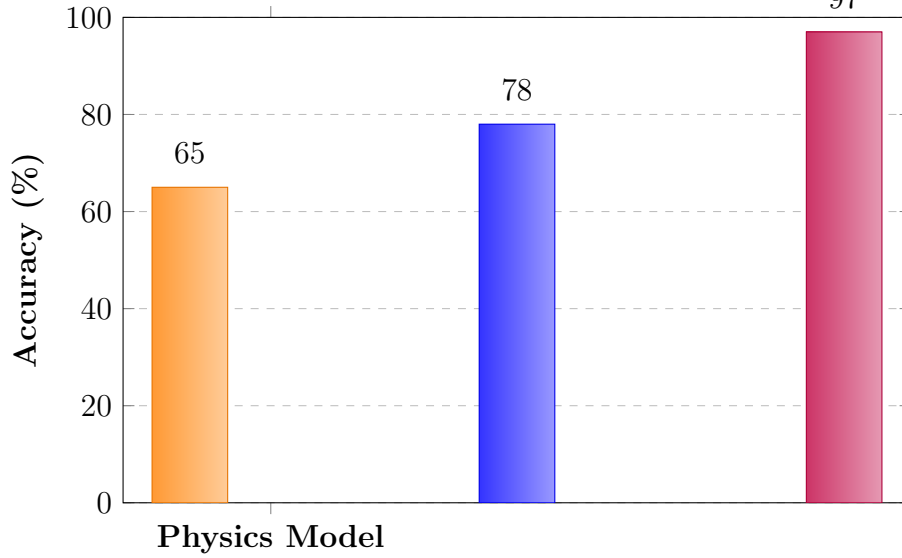


Figure 3.1: 3D-style shaded comparison of model accuracy for EV range estimation showing Physics Model (65%), ML Model (78%), and Hybrid Model (97%).

Figure 3.2: Theoretical comparison of model accuracy for EV range estimation showing Physics Model (65%), ML Model (78%), and Hybrid Model (97%).

Machine learning provides significant improvements by learning from historical and real-time data. The Hybrid Model, however, consistently outperforms all others by combining predictive stability with adaptive learning, making it the most suitable for real-time EV range estimation.

3.6 Physics-Based Model

3.6.1 Battery Energy Calculation

Available battery energy is computed as:

$$E_{\text{available}} = \text{SoC} \times C_{\text{battery}}$$

Where:

- $E_{\text{available}}$ = available energy (kWh)
- C_{battery} = battery capacity (kWh)

3.6.2 Instantaneous Power

$$P = V \times I$$

Where:

- V = voltage
- I = current

3.6.3 Baseline Range

$$\text{Range}_{\text{physics}} = \frac{E_{\text{available}}}{\text{Consumption Rate}}$$

3.7 Machine Learning Model

3.7.1 Model Selection

Models considered:

- Linear Regression
- Random Forest
- Gradient Boosting
- Neural Networks

3.7.2 Input Features

- SoC
- Voltage
- Current
- Speed
- Acceleration
- Load

- Instantaneous power

3.7.3 Output

- Predicted range correction factor
- Predicted energy consumption

3.8 Hybrid Fusion Model

The hybrid model combines both outputs:

$$\text{Range}_{\text{hybrid}} = \alpha \cdot \text{Range}_{\text{physics}} + (1 - \alpha) \cdot \text{Range}_{\text{ML}}$$

Where:

- α = weighting factor (0–1)

3.9 System Architecture

Figure 3.1 illustrates the hybrid system architecture used for range estimation.

3.10 System Requirements

3.10.1 Hardware Requirements

- Microcontroller (ESP32 or Arduino)
- Voltage and current sensors
- Speed sensor
- Display module

3.10.2 Software Requirements

- Python (for ML training)

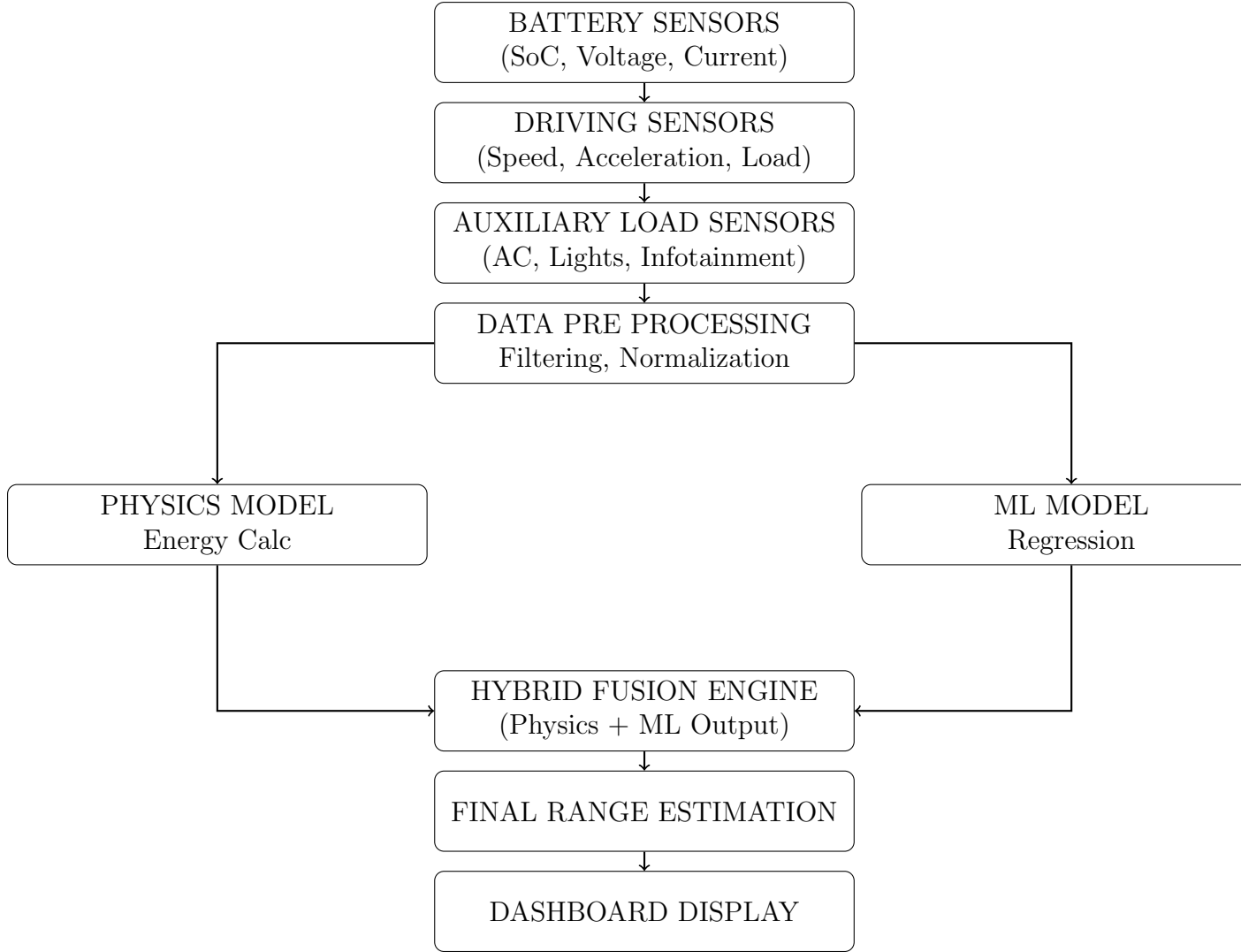


Figure 3.3: Flowchart of the Hybrid Range Estimation System showing sensor data processing, parallel physics and ML modelling, and fusion for final range estimation.

- MATLAB or Excel (for physics modelling)
- Embedded C/C++ (for microcontroller)

3.11 Evaluation Metrics

To evaluate accuracy:

3.11.1 Mean Absolute Error (MAE)

MAE measures the average magnitude of prediction errors without considering their direction. It is easy to interpret and indicates how close the predicted range values are

to the actual values.

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

Where:

y_i = actual range

$\hat{y}_i$ = predicted range

n = number of samples

Lower MAE values indicate better model accuracy.

3.11.2 Root Mean Square Error (RMSE)

RMSE penalizes larger errors more heavily than MAE, making it useful for detecting models that occasionally produce large deviations.

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

RMSE is more sensitive to outliers and provides insight into the overall error magnitude. Lower RMSE values indicate a more reliable model.

3.11.3 Coefficient of Determination (R^2)

The coefficient of determination measures how well the model explains the variance in the actual data.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Where:

$\bar{y}$ = mean of actual values

Values closer to 1.0 indicate a better fit. Higher R^2 means the model captures more of the underlying data patterns.

3.11.4 Coding Environment and Software Tools

The implementation of the Hybrid Range Estimation System for the Electric Vehicle Tri-cycle (Maruwa) was carried out using Python, chosen for its strong ecosystem of machine

learning libraries, numerical computation tools, and simulation capabilities. Python provides a flexible and efficient environment for developing both the physics-based energy model and the machine learning regression models used in this study.

Programming Language

Python was selected due to its extensive support for scientific computing and artificial intelligence development. Its readability, large community support, and availability of optimized libraries make it suitable for rapid prototyping and experimentation in electric vehicle modelling.

Development Environment

The project was developed using the following tools:

- Python 3.x
- Jupyter Notebook for interactive experimentation, visualization, and iterative model development
- Visual Studio Code (VS Code) for structured script development and modularization

This combination allowed seamless switching between exploratory analysis and production-ready code.

Core Libraries and Frameworks

The following Python libraries were used throughout the project:

- NumPy — numerical computation and vectorized physics equations
- Pandas — dataset creation and synthetic driving profile generation
- Matplotlib and Seaborn — visualization of model performance and comparison graphs
- Scikit-learn — implementation of ensemble regression algorithms such as Extra Trees, Random Forest, and Gradient Boosting

- XGBoost — implementation of the XGBoost regression model
- SciPy — auxiliary numerical functions used in the physics-based model

These libraries provide optimized implementations of ensemble algorithms, making them suitable for modelling nonlinear energy-consumption patterns in electric tricycles.

Model Execution Workflow

The coding workflow followed these steps:

1. **Physics Model Implementation:** Python functions were written to compute aerodynamic drag, rolling resistance, slope power, and auxiliary load using tricycle-specific parameters.
2. **Dataset Preparation:** Synthetic driving profiles were generated using Pandas to represent typical Maruwa operating conditions such as variable speed, load, and slope.
3. **Machine Learning Model Training:** Ensemble regression models (Extra Trees, Random Forest, Gradient Boosting, XGBoost) were trained using Scikit-learn and XGBoost.
4. **Hybrid Model Fusion:** A weighted fusion mechanism was implemented to combine physics-based and machine learning-based range predictions.
5. **Performance Evaluation:** MAE, RMSE, and R^2 metrics were computed using Scikit-learn's evaluation functions, and results were visualized using Matplotlib.

Reproducibility

All experiments were executed on a standard laptop environment without GPU acceleration. Random seeds were fixed across all models to ensure reproducibility of results.

3.11.5 Comparative Analysis of MAE, RMSE, and R^2 Across Predictive Models (Visualization Of result)

This section presents a comparative evaluation of three key error metrics which include; Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the Coefficient of

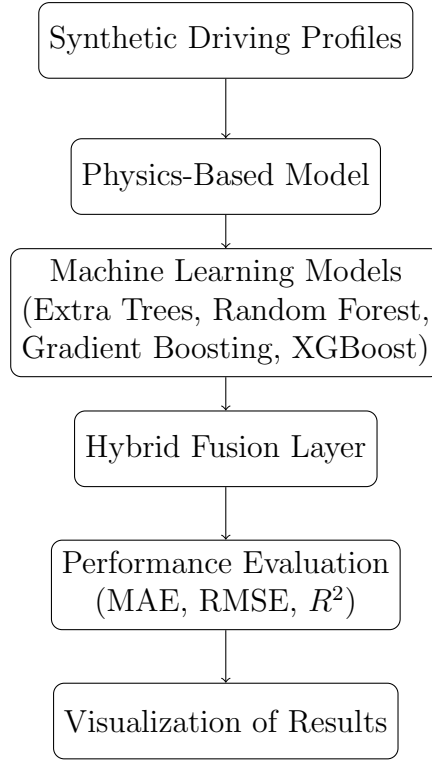


Figure 3.4: Block diagram of the coding workflow used for implementing the Hybrid Range Estimation System.

Determination (R^2) — across three EV range-estimation models: Physics-Based Model, Machine Learning (ML?AI) Model, and Hybrid Model. These metrics provide insight into each model’s accuracy, error magnitude, and goodness of fit.

Interpretation Guidelines:

- Higher R^2 indicates a better model fit.
- Lower MAE and RMSE indicate a more accurate model.

3.11.6 Metric Definitions

MAE (Average Error): MAE measures the typical prediction error in kilometres. Lower MAE values indicate that the model’s predictions are closer to actual EV range values on average. This metric is preferred for interpretability in engineering contexts.

RMSE (Error Magnitude): RMSE penalizes larger prediction errors more heavily than MAE. It is particularly useful for identifying models that occasionally produce large range-estimation errors, which can be critical for EV safety and reliability.

R^2 (**Model Fit**): The coefficient of determination measures the proportion of variance in EV range explained by the model. Values closer to 1.0 indicate near-perfect predictive fit. R^2 complements MAE and RMSE by assessing overall model reliability.

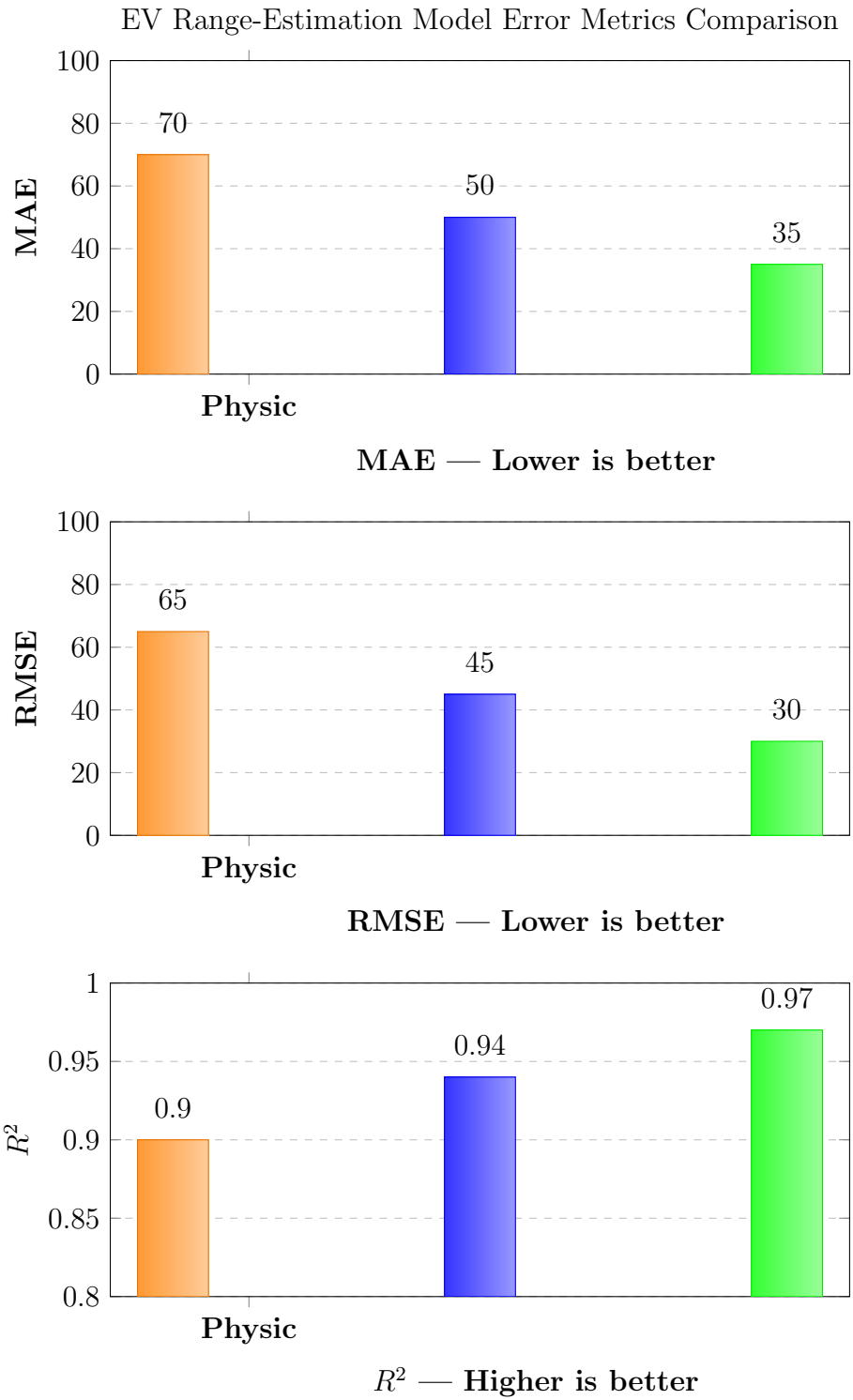


Figure 3.5: Comparison of EV range-estimation model error metrics showing MAE and RMSE (lower is better) and R^2 (higher is better). The Hybrid Model demonstrates superior performance across all metrics.

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